

(ii) **Quantum free electron theory:**

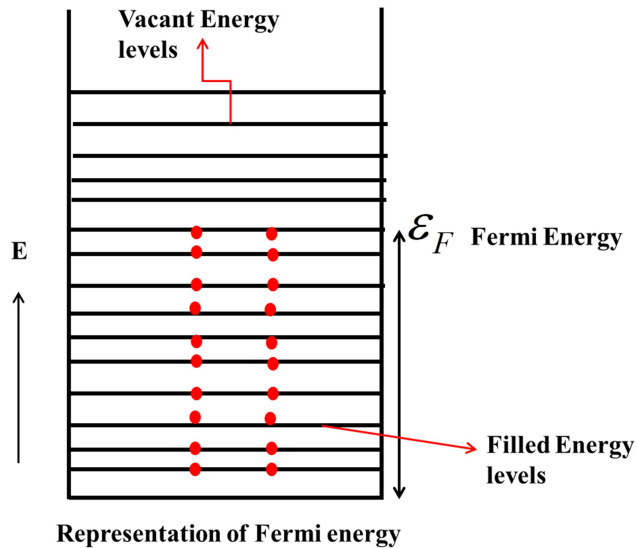
In 1928, **Sommerfeld** introduced the Quantum Free Electron Theory, wherein he posited that free electrons exhibit motion within a uniform potential. This theory adheres to the principles of quantum mechanics. The electrons in this model are treated as if they move in a potential energy well created by the positively charged atomic cores. In this potential well, electrons can have discrete energy levels, analogous to quantum states.

Assumptions of quantum free electron theory:

1. Free electrons in a metal possess discrete energy levels, which leads to energy quantization.
2. Electrons adhere to Pauli's Exclusion Principle, limiting the occupancy of any energy level to a maximum of two electrons.
3. The electron distribution across various energy levels complies with Fermi-Dirac quantum statistics.
4. Electrons within a metal generally experience consistent potential energy throughout because of the uniform potential created by ionic cores.
5. The forces of attraction between electrons and lattice ions, as well as the repulsion between electrons, are typically considered negligible.
6. Electrons are commonly described as exhibiting wave-like properties in addition to their particle-like nature.

Fermi Dirac Statistics:

1. Fermi-Dirac statistics incorporates the Pauli Exclusion Principle, which states that no two identical fermions (particles with half-integer spin) can occupy the same quantum state simultaneously. As a result, it is impossible for multiple electrons in a system to have the same set of quantum numbers.
2. Particle Distribution: Fermi-Dirac statistics describe the probability of finding particles in different energy levels. It shows that at low temperatures, particles tend to occupy the lowest available energy states, and as the temperature increases, more energy levels become populated.
3. **Fermi Energy:** *The Fermi energy ϵ_F is a critical parameter in Fermi-Dirac statistics. It represents the highest energy level occupied by electrons at absolute zero ($T = 0\text{ K}$) temperature.*



4. The distribution function defined by Fermi-Dirac statistics depends on temperature.

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

The expression given by equation (1) is also called Fermi factor.

At absolute zero ($T = 0 \text{ K}$), all energy states below the Fermi energy are occupied, and those above are unoccupied. As temperature increases, more states above the Fermi energy become occupied.

Variation of Fermi Factor or Fermi distribution function with energy and temperature:

(i) **Case I: for $\epsilon < \epsilon_F$ at $T = 0 \text{ K}$**

Value of Fermi factor

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

For $\epsilon < \epsilon_F$ and $T = 0 \text{ K}$

$$(\epsilon - \epsilon_F) < 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be negative and $\frac{1}{T} = \frac{1}{0} = \infty$.

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^{-\infty} + 1} = 1$$

$$f(\epsilon) = 1$$

Which means that $f(\epsilon) = 1$ for $\epsilon < \epsilon_F$ at $T = 0 \text{ K}$

This implies that at absolute zero ($T = 0K$) temperature, all the energy levels below fermi level are 100% occupied which is true from the definition of Fermi energy.

(ii) Case II: for $\epsilon > \epsilon_F$ at $T = 0K$

Value of Fermi factor

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

For $\epsilon > \epsilon_F$ and $T = 0K$

$$(\epsilon - \epsilon_F) > 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be positive and $\frac{1}{T} = \frac{1}{0} = \infty$.

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^{\infty} + 1} = 0$$

$$f(\epsilon) = 0$$

Which means that $f(\epsilon) = 0$ for $\epsilon > \epsilon_F$ at $T = 0K$

This implies that at absolute zero ($T = 0K$) temperature, all the energy levels above the fermi level are unoccupied or completely empty which is true from the definition of Fermi energy.

(iii) Case III: for $\epsilon = \epsilon_F$ at $T = 0K$

For $\epsilon = \epsilon_F$ and $T = 0K$

$$(\epsilon - \epsilon_F) = 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be $\frac{0}{0} = \text{Indeterminate}$

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^0 + 1} = \text{Indeterminate}$$

Which means that $f(\epsilon) = \text{undetermined}$ for $\epsilon = \epsilon_F$ at $T = 0K$

Hence, the occupation of Fermi level at $T = 0 K$ has an undetermined value ranging between zero and unity (0 & 1). The Fermi-Dirac distribution function is discontinuous for $\epsilon = \epsilon_F$ at $T = 0 K$.

(iv) Case IV: for $\epsilon = \epsilon_F$ at $T > 0K$

For $\epsilon = \epsilon_F$ and $T > 0K$

$$(\epsilon - \epsilon_F) = 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be zero

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^0 + 1} = \frac{1}{2}$$

$$f(\epsilon) = \frac{1}{2}$$

Which means that $f(\epsilon) = \frac{1}{2}$ for $\epsilon = \epsilon_F$ at $T > 0K$

For $\epsilon \ll \epsilon_F$ the probability starts decreasing from 1 and reaches 0.5 at $\epsilon = \epsilon_F$ and for $\epsilon > \epsilon_F$, it further falls off as shown in figure. In conclusion, the Fermi energy is the most probable or average energy of the electrons in a solid.

